

A Curve Fitting Approach for Non-Iterative Divider Design with Accuracy and Performance Trade-off

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Abstract—This paper presents an approach based on the curve fitting method for the design of non-iterative divider circuits with accuracy and area-delay product (ADP) trade-offs. The curved surfaces representing the quotient are partitioned into several regions, each of which is then approximated by a square/triangular plane. The planes are obtained by using the curve fitting method for accuracy optimization. The proposed architecture for implementing the planes contains only simple arithmetic operations and a look-up table. Several non-iterative divider circuits with different accuracies and ADPs are obtained. The accuracy achieved in terms of the maximum absolute error percentage (MAEP) ranges from 1.87% to 0.14%. The MAEP of 0.14% is 30% better than the one achieved by the best existing non-iterative divider.

I. INTRODUCTION

Arithmetic divider is a basic yet complex module in integrated circuits. It plays an increasingly important role in a variety of fields and is widely used in graphic processing [1]. However, divider is often a performance limiting module in these applications due to its complexity. In the VLSI implementation of digital signal processing (DSP) systems, division is also a costly operation [2], often realized by multiplication using multiply-and-accumulate (MAC) units. The lack of fast and compact division module in integrated circuits has for a long time been the bottleneck for some real-time multimedia applications, such as high performance graphics rendering pipelines [3]. The design of division circuit has received far less attention than the designs of adder and multiplier have [4], hindering the performance in many applications.

Other than those implemented with multiplications, the division circuits can be generally categorized into two main classes [4], the digit recurrence method and the functional iteration algorithm. The digit recurrence method evolves from the concept of the manual long division to obtain one or several digits each cycle, and its representation is the well-known SRT (Sweeney, Robertson, and Tocher) division method [5]. The functional iteration algorithm tries to estimate an initial approximation, and then converges to the actual result iteratively based on a functional algorithm. The Newton-Raphson algorithm [6] is a typical functional iteration method. Both types of dividers have low hardware complexity, but suffer from large latencies and/or long delay, resulting in a bottleneck in real-time applications such as 3D graphic computing [7].

We have observed that many real-time applications have a good tolerance for computation accuracy, thus leaving space for exploring the trade-off among accuracy, computation speed and hardware complexity. One such example is the 3D

graphic processing [8], where multiplications and divisions are simplified as addition/subtraction operations in logarithmic domain using the Mitchell-based logarithmic approximation method [9] [10]. For fast division, a non-iterative Mitchell-based design was proposed for the first time in [11], which achieves an accuracy of 0.20% in terms of the maximum absolute error percentage (MAEP). However, the details of the method used to obtain the approximation functions are not reported. As such, the design is unlikely to be optimized for accuracy, performance and hardware complexity. Moreover, when a divider with another accuracy and/or speed and size is needed, it cannot be obtained systematically.

In this paper, we propose a systematical approach based on curve fitting to obtain the approximation functions for the designs of fast non-iterative divider. By trading-off accuracy and area-delay product (ADP), several designs with different MAEPs ranging from 0.14% to 1.87% and hardware complexity are obtained to suit the requirements of various applications. To the best of our knowledge, there is no divider designed based on the curve fitting approach. With such divider designs, the applications that can tolerate lower accuracy can employ more compact but less accurate dividers. For those applications that demand higher accuracy, larger but more accurate dividers can be used. In the current implementations, the most accurate quotient achieved has a MAEP of 0.14%, which is 30% better than of the one achieved by the only existing non-iterative divider [11].

The proposed approximation method is described next in Section II. Section III presents the hardware architecture. The synthesis results and comparisons are presented in Section IV. Section V concludes the paper.

II. PROPOSED METHOD

A. Formulation of Division

For the purpose of proper normalization and fair performance comparison with the existing designs, the proposed divider takes as inputs two 22-bit inputs, N_a and N_b , as given in (1) and (2). For reasonable precision and dynamic range, both the inputs and the output have 1 sign bit, 5 bits of exponent and 16 bits of mantissa. It is implied that there is always a “1.” in front of the 16-bit mantissa.

$$N_a = (1 + m_a) \times 2^{e_a - 15} \quad (1)$$

and

$$N_b = (1 + m_b) \times 2^{e_b - 15} \quad (2)$$

The quotient Q of N_a divided by N_b is hence given as (3).

$$Q = \frac{N_a}{N_b} = \frac{1 + m_a}{1 + m_b} \times 2^{e_a - e_b} \quad (3)$$

If the mantissa $(1 + m_a)/(1 + m_b)$ is normalized to be in the range of $[1, 2)$, the quotient Q of (3) can be expressed as (4).

$$Q = \frac{N_a}{N_b} = \begin{cases} M \times 2^{e_a - e_b - 1}, & \text{if } m_a < m_b \\ M \times 2^{e_a - e_b}, & \text{otherwise} \end{cases} \quad (4)$$

where

$$M = \begin{cases} 2 \times \frac{1+m_a}{1+m_b}, & \text{if } m_a < m_b \\ \frac{1+m_a}{1+m_b}, & \text{otherwise} \end{cases} \quad (5)$$

is the normalized mantissa. Hence the exponent and mantissa of Q can be computed independently, other than there could be a carry from the mantissa to the exponent due to normalization.

The normalized quotient is confined within the same small range $[1, 2)$ and falls on two curved surfaces along the diagonal $m_a = m_b$ when the quotient is graphically plotted against m_a and m_b as shown in Fig. 1(a). Thus, the quotient Q can be obtained by finding the value on the surfaces corresponding to m_a and m_b . As it is complicated and costly to implement the curved surfaces, we propose to approximate them with planes as described in the next section. Fig. 1(b) shows an example of percentage error distribution when the curved quotient surface is approximated by 16×16 planes.

B. Approximation of Division

The proposed approach is based on the curve fitting method to approximate the curved surfaces with small square planes for efficient hardware implementation. The original curved surfaces are first evenly partitioned into certain number of regions. The curved surface in each region is then approximated with a plane represented by (6).

$$\hat{M} = 1 + p \times m_a - q \times m_b + r \quad (6)$$

where $\hat{M}$ represents the approximated M in (5), and p , q and r are the constant coefficients within each partitioned region to form the corresponding plane.

It is worth mentioning that the quotient surfaces in Fig. 1(a) are disjoint along the diagonal where $m_a = m_b$. Large approximation error may be resulted in these regions if square planes are used to approximate them, because the approximated mantissa may fall on the totally wrong location and the exponent may also be wrongly computed due to wrong carry bit. As such, each square region along the diagonal is further divided into two triangular regions, each approximated by a triangular plane. The triangular planes are represented by the same equation (6). The hardware implementation of (6) is straightforward and efficient as only simple arithmetic modules such as shifters, adders/subtractors and look-up tables are required.

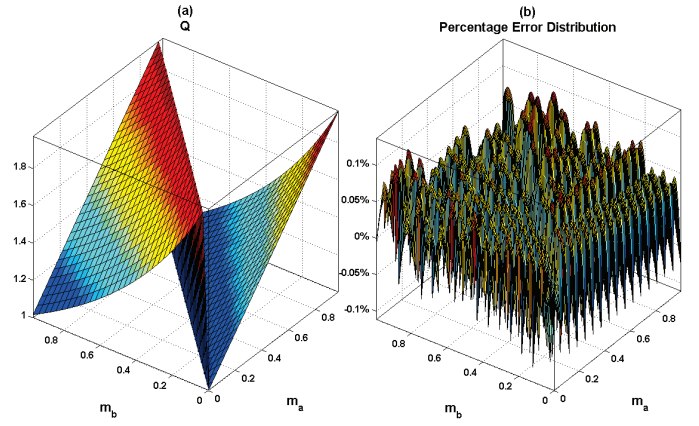


Fig. 1. (a) Normalized division curved surfaces. (b) Percentage error distribution when approximated by planes.

As the original curved surfaces are approximated by small square/triangular planes, there are approximation errors, leading to loss of accuracy. A design is acceptable as long as the approximation errors are within the accuracy tolerance specified in an application. The approximation error, η , in terms of the MAEP can be derived from (4), (5) and (6) as given in (7), which shows that η is independent of the exponent.

$$\eta = \left| \frac{\hat{Q} - Q}{Q} \right| = \left| \frac{\hat{M} - M}{M} \right| \times 100\% \quad (7)$$

where $\hat{Q}$ denotes the approximated quotient of Q .

The accuracy and the complexity of the design are impacted by several parameters, specifically, the number of planes used for the approximation, the values of the constant coefficients p , q and r , as well as their wordlengths. It is possible to obtain designs that are more accurate but complex, or simpler but less accurate, providing a trade-off of accuracy and ADP. In this work, the parameters are obtained as described below.

We first decide the partition scheme and adopt the curve fitting approach to find the coefficients p , q and r for the square/triangular plane to approximate the individual region of the curved surfaces. During the curve fitting, uniform sampling points are selected from each partitioned curved surface. The function `[za, goodness] = fit([ma, mb], z0, 'poly11')` from the Matlab curve-fitting toolbox is then used to find a two-variable expression that best fits all the sampling points, and the approximation error for each region is then calculated. In the function, `za` and `goodness` are respectively the returned fitting function, i.e. equation (6) in this case, and the fitting quality. `ma` and `mb` are the sets of sampling points and `z0` is the set of values of the function corresponding to the sampling points. Type `'poly11'` indicates that the fitted function type is a linear polynomial surface. We have experimented using different number of sampling points for the fitting and found that the approximation error varies negligibly when the sampling points are more than 100 points. The coefficients p , q and r for each region are obtained after the curve fitting.

We also experimented with two schemes of partitioning the curved surface, an 8×8 and a 16×16 partitions, each with

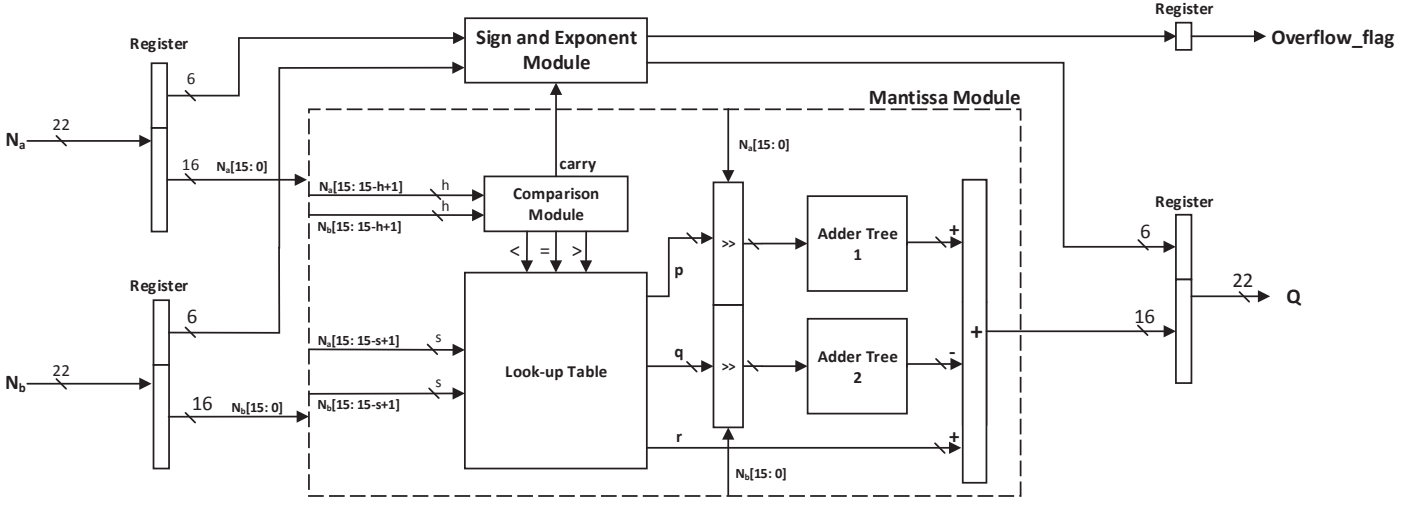


Fig. 2. Hardware architecture for the proposed non-iterative divider

different wordlength of the constants, and obtained several divider designs with different accuracies and ADPs. The detailed results, including the wordlengths of the coefficients, the areas and delays are presented in Section IV.

III. PROPOSED HARDWARE ARCHITECTURE

Based on the above approach, the architecture of the non-iterative divider is proposed as shown in Fig. 2. The main functional modules are elaborated below.

A. Mantissa Module

The mantissa module takes as inputs the 16-bit mantissas of N_a and N_b and produces the 16-bit mantissa of their quotient Q . It consists of a comparison module, a look-up table, two shift modules, two adder-tree modules and a summation module.

1) *Comparison Module*: The module compares h MSBs (Most Significant Bits) of the mantissa. h is determined after the coefficients p , q and r are found and the overall approximation error is calculated. It is chosen as the smallest integer that satisfies $2^{-h} < \text{error}$, as any values smaller than 2^{-h} are insignificant and can be ignored for hardware simplicity. The comparison results, $m_a < m_b$, $m_a = m_b$ or $m_a > m_b$, determine whether a square plane is on the diagonal and if so, on which triangular plane (see the situations discussed in Section II). The module also produces a carry for the Sign and Exponent Module.

2) *Look-up Table*: The Look-up Table stores the coefficients p , q and r . s MSBs of the mantissa are used to read the coefficients, where $s = 3$ and 4 respectively for the 8×8 and 16×16 partition scheme. The comparison results from the Comparison Module are used for accessing the coefficients of the triangular planes if a square plane is on the diagonal.

3) *Shift Modules and Adder Trees*: The module is to produce the partial products of $p \times m_a$ and $q \times m_b$ of (6). The mantissas m_a and m_b are first shifted using hardwired shifters for all the required shifted positions and multiplexed. The shifted values are selected by the corresponding non-zero bits of p and q to be summed by the carry-save adder trees.

4) *Summation Module*: The results of the adder trees and the coefficient r are summed in this module to produce the approximated mantissa of the quotient in (6).

B. Sign and Exponent Module with Overflow Indication

The module generates the sign bit and the 5-bit exponent for the quotient. The sign bit is simply realized by an XOR operation of the two input sign bits. The exponent is produced by a 5-bit subtractor, performing $e^a - e^b + 15 - \text{carry}$. Note that the exponent is not approximated.

The module also generates an overflow flag if the resulting exponent of the quotient Q is out of the valid range of $[-15, 16]$.

IV. PERFORMANCE EVALUATION AND COMPARISON

There are only a couple of existing non-iterative divider circuits [12] [11] and both of them perform the division as subtraction in the logarithmic domain. The proposed approach uses the curve-fitting method to systematically obtain several non-iterative divider circuits with different accuracy and ADP trade-offs as listed in Table I, where the number of partitions, the wordlengths of the coefficients, the MAEPs achieved and the synthesis results are included. The synthesis results are obtained from coding the proposed architecture of Fig. 2 in Verilog RTL for synthesis in 90 nm CMOS process with the Synopsys Design Compiler. The MAEP achieved by the proposed approach ranges from 1.87% to 0.14%. The circuit with the best MAEP is obtained by partitioning the curved surfaces of the quotient into 16×16 square planes. It is worth mentioning that the MAEP in Table I is reduced when the number of partitions and the wordlength of p , q and r are increased, resulting in larger ADPs. On the other hand, increasing the wordlength of the input mantissas does not help in reducing the MAEP. The wordlength of the input exponents has no impact on the MAEP but marginally on only the area.

Table II compares the two most accurate designs, i.e. the last two designs in Table I, with the existing non-iterative dividers reported in [11]. Compared with the single-stage method [11], Design A and B have a lower MAEP by 20% and

TABLE I. DIVIDER CIRCUITS WITH MAEP AND ADP TRADE-OFF

Partition Planes	Wordlength (bits)		h	MAEP	Delay (ns)	Area (μm^2)	ADP
	p and q	r					
8×8 (s=3)	6	6	6	1.87%	2.44	4896.5	11974
	7	10	7	0.88%	2.50	5054.4	12636
	8	10	8	0.67%	2.59	5240.9	13574
	9	10	8	0.62%	2.65	5380.4	14258
16×16 (s=4)	8	12	8	0.56%	2.63	9846.5	25896
	9	12	9	0.25%	2.69	10596.2	28504
	10	12	10	0.16%	2.72	11183.4	30148
	11	13	10	0.14%	2.85	12324.9	35126

TABLE II. PERFORMANCE COMPARISON WITH PREVIOUS WORKS

Designs	MAEP	Delay (ns)	Area (μm^2)	ADP
3-stage method [12]	0.21%	5.22	8939	46662
Single-stage method [11]	0.20%	2.85	10680	30438
This work – Design A	0.16%	2.72	11183	30418
This work – Design B	0.14%	2.85	12325	35126

30% respectively. Design A has shorter delay and comparable ADP while Design B has the same delay but larger ADP.

The distribution of the MAEP and ADP of the eight designs obtained can be visualized from Fig. 3. It is observed that the 8×8 and 16×16 partitions have different design spaces. With the former, lower hardware complexity and shorter delay due to smaller look-up table and simpler adder trees are achieved, but the accuracy is limited. With the latter, higher accuracy is achieved at the expense of higher complexity and longer delay. Hence, a series of divider designs with different accuracy-ADP trade-offs are obtained for various applications where fast division is needed in integrated circuits.

In the future, work can be done to obtain divider circuits with other accuracies and ADPs by adopting other partition schemes, including irregular partition schemes. The implementation of the proposed architecture can also be further optimized by merging the adder trees and the summation module in the mantissa module.

V. CONCLUSION

An approach based on the curve fitting method is proposed for non-iterative divider designs. The curved surfaces representing the quotients are approximated by several square/triangular planes. The constant coefficients for the plane equations are obtained by the curve fitting method to optimize the accuracy. A series of designs with accuracy and ADP trade-offs are obtained, providing various solutions for applications that have different complexity and accuracy requirements. With the systematical curve fitting method, the accuracy achieved in terms of MAEP ranges from 1.87% to 0.14%, where the MAEP of 0.14% is 30% better than the best existing accuracy achieved by the non-iterative design reported in [11].

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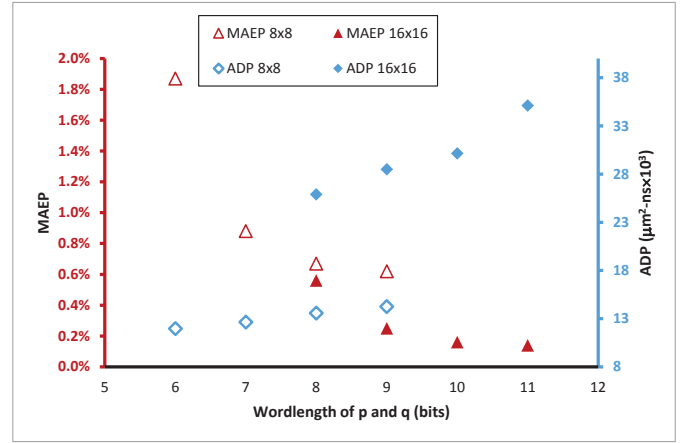


Fig. 3. MAEP and ADP trade-off for various partition scheme and coefficients wordlengths

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